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Speciation of the reaction intermediates from *n*-dodecane oxidation in the low temperature regime

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Abstract

The oxidation of *n*-dodecane was studied with experiments in a pressurized flow reactor over the low temperature regime (550–830 K) at elevated pressure (0.8106 MPa) and lean equivalence ratio (0.23). *n*-Dodecane exhibited negative temperature coefficient (NTC) behavior such that fuel reactivity peaked around 700 K and then decreased with increasing temperature. Samples were extracted at a residence time of 120 ms and stable intermediates were analyzed utilizing a gas chromatograph with a flame ionization detector coupled to a mass spectrometer. While a range of species were observed, a significant portion of *n*-dodecane decomposition was attributed to a select few species. These major carbon contributing intermediates included carbon oxides (CO and CO_2), light aldehydes (formaldehyde, acetaldehyde, and propanal), C_{12} alkylated tetrahydrofurans (2-propyl-5-iso-pentyltetrahydrofuran and two other incompletely identified isomers), lactones (5-methyl-dihydro-2(3*H*)-furanone and dihydro-2(3*H*)-furanone), and alkenes (ethene and 3-dodecene). The experimental results were compared to two kinetic models. The models captured the NTC behavior; however, discrepancies in measured formaldehyde, several light unsaturated species, and carbon monoxide indicate areas for model improvement. In the low temperature branching pathway from Livermore, a reaction path was identified (hydrogen abstraction from a secondary carbon followed by molecular oxygen addition and then internal hydrogen shift from the primary carbon) which would, if more active, predict the measured intermediate species. The data from this study provide targets for the continued refinement of kinetic models for *n*-dodecane.
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1. Introduction

The United States Department of Defense Directive 4140.43 mandated the use of JP-8 jet fuel as the universal military fuel where applicable [1]. It is desirable to optimize fuel economy, power output, fuel flexibility, and emissions for all power systems fueled with JP-8. In the evolution of advanced engines, simulations are becoming preferable to testing in developing systems. However, JP-8 is a complex fuel that contains hundreds of components, many with unknown chemistry,

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and proper simulations require resources beyond current computational abilities. To reduce these requirements, the use of surrogate fuels, mixtures of approximately 1–10 components that mimic the properties and behaviors of real fuels, has been recognized as a feasible approach for predicting combustion behavior [2]. *n*-Dodecane is a likely major component of JP-8 surrogate fuels and is the target of this study.

While analytical models of individual reactions have become more useful in developing chemical kinetic mechanisms, experimental measurements remain the benchmark for determining reaction pathways. Experimental targets such as burning velocities in flames, ignition delays in rapid compression machines (RCM) and shock tubes, and reactivity tracer species in flow reactors have been extensively used. Chemical samples from the oxidation of C_7–C_{10} alkanes in RCM's and flow and jet-stirred reactors have been analyzed for intermediate species concentrations [e.g., 3–9]. Such measurements provide fundamental insights to the key reactions in a mechanism. Hanson and co-workers measured OH concentrations from *n*-dodecane combustion behind reflected shock waves in a shock tube [10]. Below 900 K, the negative temperature coefficient (NTC) regime was observed where reactivity decreases with increasing temperature. Gomez and co-workers measured intermediate species from methane and ethene counterflow diffusion flames doped with a JP-8 surrogate containing 30% *n*-dodecane, with temperatures in the flame ranging from 400 to 2000 K [11,12]. Major intermediates included light alkenes and alkanes. Dagaut and co-workers measured intermediates from *n*-hexadecane in a jet-stirred reactor at temperatures of 1000 K and higher [13]. Fournet and co-workers measured intermediates, including alkenes, alkynes, dienes, and aromatics, from *n*-dodecane pyrolysis in flow and jet-stirred reactors [14,15]. The difficulties of making high quality measurements during combustion of these high molecular weight hydrocarbons, due to a combination of low volatility and high reactivity, have long been documented [16].

The present study focused on the oxidation of *n*-dodecane over the low temperature regime of hydrocarbon oxidation, which is particularly important in autoignition in combustion systems. The general pathways of the oxidation of lighter *n*-alkanes, such as *n*-heptane, have been evaluated in several experimental studies and the results have been used to validate detailed kinetic models [17–19]. The key pathways should be applicable for heavier *n*-alkanes, and thus a brief summary of low temperature oxidation of *n*-alkanes is included to stress the major pathways; more detail can be found in recent reviews [20–23]. In the critical branching pathway for low temperature oxidation, the alkane undergoes hydrogen abstraction to yield an alkyl radical, which undergoes molecular oxygen addition to produce the alkylperoxy radical. This isomerizes to form the alkylhydroperoxy radical. Further molecular oxygen addition produces the peroxyalkylhydroperoxy radical. Decomposition produces a hydroxyl radical and a ketohydroperoxide, which decomposes to produce a second hydroxyl radical and other species. The higher end of the low temperature regime is the NTC region. In this temperature regime, instead of following pathways to produce the highly reactive hydroxyl radical, the reaction proceeds to produce less reactive species including hydroperoxy radicals and alkenes. At even higher temperatures, moving through the NTC region and into the intermediate temperature regime, pathways lead from the hydroperoxy radical to hydrogen peroxide, which at high enough temperatures decomposes to form two hydroxyl radicals leading to hot ignition. While much is known, low temperature models are still under refinement.

2. Methodology

The pressurized flow reactor (PFR) facility at Drexel University was used to conduct the experiments. The PFR is a turbulent flow reactor designed to study the chemistry of hydrocarbon oxidation, in relative isolation from fluid mechanics and large temperature gradients. As detailed descriptions can be found elsewhere [24,25], the facility and methodology will only be summarized here; however, modifications to accommodate high molecular weight hydrocarbons will be described in detail. Inside the PFR, the reaction chamber consists of a 2.2 cm I.D., 40 cm long quartz reactor tube within a pressure vessel. The pressure vessel and the inside of the reactor tube are separated by an annulus that is at the same pressure as the reaction chamber. At one end of the reactor tube is the fuel mixing nozzle and the other end contains the entrance for the gas sampling probe and an integral type-K sample thermocouple. Nitrogen (purity = 99.9%) and oxygen (purity = 99.994%) are mixed to form a synthetic air free of contaminants. This synthetic air is heated to the reaction temperature. To maximize temperature uniformity along the quartz reactor tube within the pressure vessel of the PFR, experiments are conducted at high dilution in nitrogen to limit the temperature rise due to heat release and the walls of the pressure vessel are heated by three independently controlled heaters. For the present experiments, the temperature rise due to heat release varied from 11 K at maximum reactivity to 0 K at no reactivity.

The liquid fuel, *n*-dodecane (purity = 99%), was injected into the centerline of a heated nitrogen stream one meter from the reactor inlet to ensure complete vaporization and mixing. The

synthetic air and the prevaporized fuel/nitrogen were rapidly mixed (<1.5 ms) in an opposed jet annular mixing nozzle at the entrance of the quartz reactor tube. At the other end, a water-cooled, borosilicate glass-lined stainless steel probe extracted samples from the centerline of the quartz reactor tube and quenched the chemical reactions. The residence time is held constant by adjusting the position of the sampling probe, and the pressure is held constant by adjusting the pressure regulating valve.

Previously in the PFR facility, a sample storage cart, consisting of an oven containing stainless steel sample loops, was used for the storage of samples. While the process was adequate for lighter hydrocarbons, the sample storage for *n*-dodecane led to unacceptable carbon balances. In particular, condensation of heavy species and catalytic reactions with the stainless steel surfaces led to a loss of gas-phase species. Thus, a direct transfer controlled cool down (DT-CCD) facility and methodology was developed to reduce sample collection-to-analysis times (from up to 24 h previously to 180 s now). The analytical chemistry facility consisted of a gas chromatograph with a flame ionization detector coupled to a mass spectrometer (GC/MS/FID). The GC/MS/FID was moved to a few meters from the exit of the PFR facility so that samples could be safely directed from the PFR to the GC. During the DT-CCD, the sample temperature is allowed to cool and re-stabilized at each sampling point by adjusting the heaters. Samples continuously flow at 0.025 L/s from the exit of the PFR through a heated borosilicate glass-lined stainless steel tube to a heated three-way valve. From the three-way valve, the flow is sent to a nondispersive infrared (NDIR) analyzer for carbon monoxide (CO) and carbon dioxide (CO_2) measurements (±25 ppm uncertainty) and an electrochemical cell for oxygen (O_2) measurements (±1250 ppm uncertainty). When a sample is desired for detailed measurements, the sample temperature is held constant so that CO, CO_2, and O_2 levels remain steady. The three-way valve is then adjusted so that the sample rapidly fills a heated, evacuated 1-mL stainless steel sample loop to 76 KPa. Monitoring the pressure of the sealed sample loop confirmed that no leaks were present. This injection pressure maintained consistency with previous PFR experiments using the sample storage cart, where the select pressure allowed for a repeat injection if the first analysis failed. The sample is then injected onto the column. The quenching time, defined as lowering the sample below 473 K, is 1 ms. Sample transfer time after quenching to fill the 1-mL sample loop, is <1 s. The GC pre-injection time, from the 1-mL sample loop, through an internal heated transfer line, and to the column, is 180 s. To determine the initial fuel concentration, a fuel calibration sample (fuel and N_2) is extracted at a temperature above the NTC region where the low temperature reactivity has ceased (typically 835 K) and directed to the GC/MS/FID after flowing through the PFR. The GC chromatogram from the fuel calibration sample showed no additional species, confirming fuel stability during the sampling process.

For species separation, the GC method utilized a 100-m polydimethylsiloxane-based capillary column and a GC oven temperature program of 253–523 K for separation of all species. For identification, the MS operated under electron ionization mode. Mass spectra were compared to the NIST MS 2.0 database, which contains 150,000 mass spectra from worldwide contributors [26]. Identification was reinforced by duplicate spectra from different contributors to the database. For supplementary identification, GC retention times were matched to standards, where available. For quantification, calibration curves were established with standards of gas-phase hydrocarbons at 15, 100, and 1000 ppm. Other species were quantified with correction factors of FID signals that accounted for differences in carbon, hydrogen, and oxygen number between different compounds of similar structure [27]. Formaldehyde was identified with the MS. The linear relationship observed between the MS and FID signals for acetaldehyde was used to determine a theoretical formaldehyde FID signal from the formaldehyde MS signal.

Three replicate DT-CCD experiments were conducted; all input parameters and results were averaged. The initial molar fraction of *n*-dodecane was an average of 531 ppm with one standard deviation of ±8 ppm. The intended fraction was 530 ppm to limit heat release. The initial oxygen molar fraction was 42,100 ppm, with the remaining initial balance being nitrogen. This was an equivalence ratio of 0.23 ± 0.05. All experiments were conducted at 0.8106 ± 0.0025 MPa pressure and 120 ± 10 ms residence time. For CO, CO_2, and O_2, experimental results were recorded at <5 K intervals from 550 to 830 K. The data are therefore a map of reaction progress at varying temperature while inlet concentrations (fuel, oxidizer, and diluent), residence time, and pressure are maintained constant. During each DT-CCD process, samples were extracted and stable intermediates were measured at sample temperatures of 550, 640, 700, 750, and 830 K, with an uncertainty of ±2 K. In the experimental results reported in Section 3, the concentrations are the average of the three experiments, with error bars considered to be ±1 standard deviation.

Two models, the Lawrence Livermore National Laboratory (Livermore) and the Politecnico di Milano (Milano), were evaluated for comparison to the *n*-dodecane experiments. The Livermore model is a detailed mechanism describing the oxidation and pyrolysis of C_8–C_{16}

n-alkanes [28]; and the Milano model is a semi-detailed model for the pyrolysis, partial oxidation, and combustion of hydrocarbons up to C_{16} [29]. Both models have the critical low temperature branching pathway as described in Section 1, as well as other paths (alkyl radical decomposition to form alkene and smaller alkyl, and alkylhydroperoxy decomposition to form either alkene and hydroperoxy radical, or cyclic ether and hydroxyl radical, or alkene, ketene, and hydroxyl). The key difference from Livermore is that the Milano model lumps isomers. Consequently, the Livermore model has reactions describing C_{12} ketohydroperoxide decomposition depending on the isomer, while the Milano model has three independent global C_{12} ketohydroperoxide decomposition reactions. Livermore includes C_{12} alkylated tetrahydrofurans while Milano includes only lighter cyclic ethers. Another *n*-dodecane model was evaluated but the results are not presented here as the model is applicable only for high temperatures [30]. Each model, and its associated thermochemical data file, was used without modification, the purpose being to identify potential areas for improvement. The models were exercised using the plug flow reactor module at adiabatic conditions with CHEMKIN-PRO [31] which approximates the operation methodology employed for this study. At their respective peak reactivities, the Livermore and Milano models predicted 23 K and 20 K temperature rises; temperature rises decreased with decreasing reactivity. No surface reactions were included in the model as they were demonstrated to be unimportant in this reactor [24]; the absence of reactivity observed at 830 K indicates that oxidation is not influenced by the surface. Simulations were run at 10 K intervals from 520 to 850 K to cover the temperature range of the experiments, and the input parameters were the averages of the experiments.

A rate-of-production (ROP) analysis was performed for the following species experimentally observed (*n*-dodecane, CO, CO_2, formaldehyde, and C_{12} alkylated cyclic ethers). Both the Livermore and Milano models were analyzed to isolate the key reactions that produce the major species. The analysis was conducted at sample temperatures of 640, 700, 750, and 830 K. As intermediate species were not measured in significant concentrations at 550 K, no ROP analysis was conducted at that temperature. Only reactions with ROP's of at least ±3% of the net ROP of the species were considered significant for this analysis. In Section 4, reaction numbers refer to the sequence in the input files obtained from Livermore (L) and Milano (M).

3. Results

For the experiments reported, carbon balances were 80–110%, and oxygen balances were 97–101%. One hundred fifteen species were identified and measured; however, many of them were at trace amounts. Detailed speciation data for all species measured can be found in the Supplementary data. Summing only a few species accounted for 55–80% of the initial carbon, depending on the temperature. These species were CO, CO_2, formaldehyde, acetaldehyde, propanal, 2-propyl-5-iso-pentyltetrahydrofuran and two other incompletely identified C_{12} alkylated tetrahydrofuran isomers, 5-methyl-dihydro-2(3*H*)-furanone, dihydro-2(3*H*)-furanone, ethene, and 3-dodecene. Additional aldehydes, cyclic ethers, lactones, alkenes, ketones, diones, and carboxylic acids were measured at smaller carbon contribution quantities.

Figure 1 presents the experimental and modeling results of *n*-dodecane and CO in carbon molar fraction, with these two species showing the expected inverse relationship. In Figs. 1–5, the sample temperature refers to the temperature of the products at the 120 ms residence time sampling location in the quartz reactor and thus includes temperature rise due to heat release in experiments and models. CO accounted for up to 10% of the carbon. The

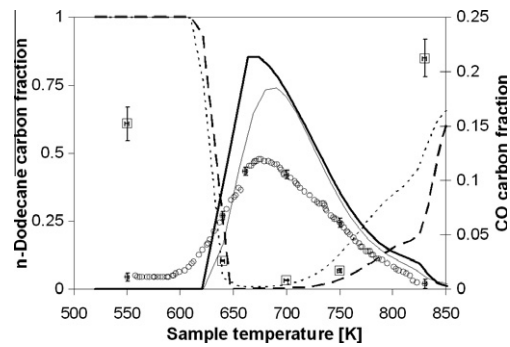

Fig. 1. Carbon fractions for *n*-dodecane: (□) experimental, (———) Livermore, (- - - - - - -) Milano; and for CO: (○) experimental, (—) Livermore, (——) Milano.

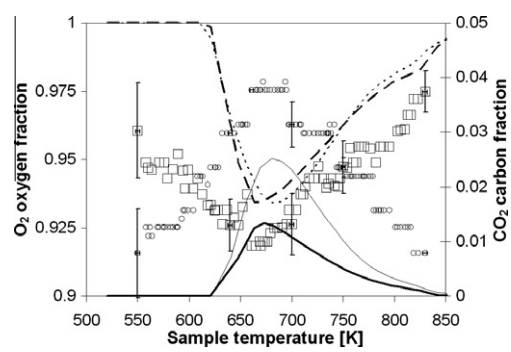

Fig. 2. Oxygen fractions for O_2: (□) experimental, (———) Livermore, (- - - - - - -) Milano; and carbon fractions for CO_2: (○) experimental, (—) Livermore, (——) Milano.

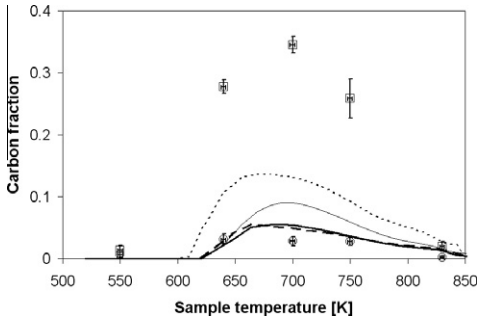

Fig. 3. Carbon fractions for formaldehyde: (□) experimental, (—) Livermore, (——) Milano; and for acetaldehyde plus propanal: (○) experimental, (— — —) Livermore, (- - - - - - -) Milano.

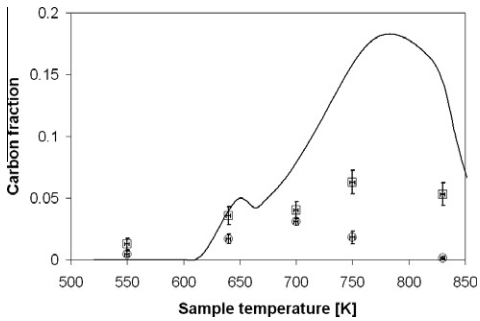

Fig. 4. Carbon fractions for C_{12} alkylated tetrahydrofurans: (□) experimental, (——) Livermore; and for 5-methyl-dihydro-2(3*H*)-furanone: (○) experimental.

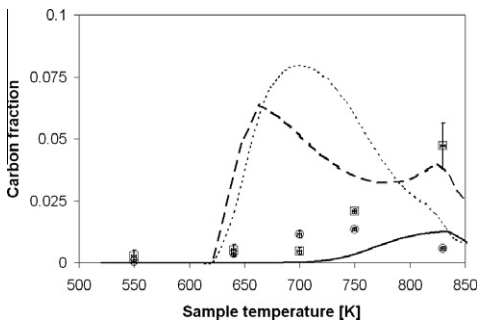

Fig. 5. Carbon fractions for 3-dodecene: (□) experimental, (—) Livermore; and for ethene: (○) experimental, (— — —) Livermore, (- - - - - - -) Milano.

smooth profile of CO (the main indicator of reactivity) suggests that similar behavior can be expected for all species. Both the Livermore and Milano models captured the general low temperature reactivity trend. In the low temperature regime, CO is a good indicator of reactivity and does not oxidize significantly to CO_2 [32]. The maximum CO, 760 ppm, occurred at 675 K. This is traditionally considered the start of NTC behavior. The Livermore and Milano models over predicted peak CO production by approximately 2 and 1.5 times, respectively.

The consumption of O_2 is shown in Fig. 2, in terms of overall oxygen fraction; as expected O_2 closely tracks the fuel consumption and CO production. Since the models do not predict any *n*-dodecane consumption below 610 K, the models correspondingly do not predict any O_2 consumption at the lower temperatures. Above 640 K, however, the model predictions are good. The models under predict the consumption of O_2 at the higher temperatures, but considering the experimental uncertainty, the experiments and predictions are close. In general, because of the lean equivalence ratio, O_2 is not a controlling factor. As such, only up to 8% of the O_2 is consumed.

Figure 2 also plots the carbon contribution from CO_2. The trend of CO_2 followed the CO trend, but at approximately 1/3 the quantity. At low temperatures, CO_2 can be produced from oxygenates such as cyclic ethers [33] rather than from the oxidation of CO. Experimentally, the CO_2 maximum was 250 ppm, at 694 K. Again, both models do not predict CO_2 formation at the lower temperatures, although they capture the trend once the reaction is initiated. The Livermore and Milano models under predicted the experimental results by factors of 3 and 2, respectively.

Figure 3 displays the carbon contribution from the most dominant aldehydes, formaldehyde, acetaldehyde, and propanal, along with the corresponding model predictions. Formaldehyde was not only the dominant aldehyde measured, but was also the dominant intermediate, accounting for the majority of the carbon in the 640, 700, and 750 K samples. A formaldehyde molar fraction of 2200 ppm was measured at 700 K, although a slightly higher maximum would be expected to occur about 25 K lower at the NTC start temperature. At 700 K, the Milano model predicted a formaldehyde level four times lower, and the Livermore model predicted a level six times lower. Like formaldehyde, acetaldehyde and propanal exhibited an increase between 640 and 750 K. The Livermore model predicted the profile well, while the Milano model predicted a level five times higher.

One C_{12} alkylated tetrahydrofuran, 2-propyl-5-iso-pentyltetrahydrofuran, was positively identified and two other C_{12} alkylated tetrahydrofurans were identified, but no MS standards exist, preventing positive identification of each isomer. Figure 4 shows the measured, combined carbon contribution of these three species along with the predictions of the Livermore model that included five C_{12} alkylated tetrahydrofuran isomers. The model captured the experimentally measured trend of gradually increasing levels until near the

end of the NTC region, although the predicted levels were up to three times higher.

Several lactones were measured and Fig. 4 includes the measured carbon contribution of the dominant lactone, 5-methyl-dihydro-2(3*H*)-furanone. The measured trend is similar to that of the aldehydes with a carbon contribution roughly the same as acetaldehyde. Neither model included the production of lactones. To ensure the lactones were produced from the chemical reactions in the PFR and not from reactions occurring inside the column of the GC oven during the low temperature phase of the GC method, a validation CCD experiment was conducted with an initial GC oven temperature of 353 K. The lactones and other high molecular weight oxygenated species were still measured in similar concentrations although the GC retention times had been reduced by 30 s. Also, in motored engine experiments studying the autoignition of tetrahydrofuran at an intake temperature of 400 K, dihydro-2(3*H*)-furanone was measured [34]. Thus, we conclude that the lactones are produced in the PFR and are not an artifact of the sampling and analytical system.

Figure 5 shows the carbon contribution from the two dominant alkenes, ethene and 3-dodecene. Only the Livermore model, which includes six isomers, includes dodecene. It captured the trend of increasing 3-dodecene with temperature, although peak production was under predicted by a factor of four times. Ethene peaked at 750 K and, while both models captured the trend, levels were over predicted.

4. Discussion

The ROP analysis was conducted to provide insight into the dominant reaction paths and the differences between the experiments and models. For both models, the initial reaction is hydrogen abstraction from *n*-dodecane via a hydroxyl radical. The Livermore model considers the yields for 2-, 3-, 4-, 5-, and 6-dodecyl radicals (L3144, L3145, L3146, L3147, and L3148, respectively) to be equally dominant, followed by half the production of one of those for 1-dodecyl radical (L3143), Supplementary Fig. S1 (Figs. S1–S18 can be found in the Supplementary data). The Milano model, because it is semi-detailed, lumps all these reactions as one reaction to produce one *n*-dodecyl radical (M7312), Fig. S2.

Only the detailed Livermore model included C_{12} alkylated cyclic ethers. At 750 K, numerous isomers were produced by the decomposition of various alkylhydroperoxy radical isomers to yield the C_{12} alkylated cyclic ether isomer and a hydroxyl radical (L5883, L5885, L5886, L5887, L5888, L5889, L5890, L5965, L5967, L5968, L5969, L5970, L5971, L5972, and L5973), Figs. S3–S12. At 830 K, the reduction in C_{12} alkylated cyclic ethers was partially attributed to the reversibility of those reactions. Additionally, a competing pathway with the same reactants (C_{12} alkylated cyclic ether and hydroxyl radical) was significant at 830 K and yielded an alkene, an oxygenated radical, and H_2O (e.g., L7681 and L7682 in Fig. S4).

As mentioned in Section 3, CO_2 formation via CO oxidation is not considered the primary source for CO_2 in the low temperature regime. However, in the Livermore model in Fig. S13, this reaction (L7) accounted for a significant contribution of CO_2 (producing 23% of the net CO_2 level at 700 K). Nevertheless, the dominant reaction was the oxidation of the ethynyloxy radical (HC_2O) to produce CO_2 and a formyl radical (L1454). This reaction produced 38% of the net CO_2 level at 700 K. The Milano model suggests a different key reaction pathway for CO_2 production, Fig. S14. At 700 K, the dominant reaction, contributing 38% of the net CO_2 level, is the reaction of formaldehyde with a hydroxyl radical to yield CO_2, H, and H_2 (M181). Clearly, to understand the formation of CO_2 in the low temperature regime from these large hydrocarbons, additional work is required.

For the production of CO in the Milano and Livermore models, seven and eight reactions, respectively, contributed substantially, Figs. S15 and S16. The dominant reaction was the production of CO and a hydroperoxy radical from the reaction of O_2 with a formyl radical. At 700 K, for both models, this reaction produced 50% of the CO. A wider range of significant reactions was identified for the production of formaldehyde, the dominant intermediate. Figure S17 plots the formaldehyde ROP's of the significant reactions at the different temperatures analyzed in the Milano model. The consumption of formaldehyde by reaction with a hydroxyl radical (M2533) offset the numerous reactions producing formaldehyde. A similar observation is observed in the ROP analysis for formaldehyde in the Livermore model, Fig. S18. The same reaction (L32) offset the production of formaldehyde.

The discrepancy between the measured and predicted formaldehyde (at 700 K, 2200 ppm was observed, while Livermore predicted 322 ppm and Milano predicted 543 ppm) may relate to other results. The Livermore model also predicted higher levels of CO, ethene, propene, 1-butene, 1-pentene, 1-hexene, 1-heptene, 1-octene, 1-nonene, 1-decene, 2-propenal, propanal, butanal, pentanal, hexanal, heptanal, octanal, nonanal, and C_{12} alkylated cyclic ethers. Other species predicted by Livermore (at least 10 ppm), but not measured experimentally (detectability is 1 ppm), included decanal, ketene, methyl peroxide, formic acid, and 1,3-butadiene. The Milano model predicted higher levels of CO, ethene, propene, *n*-butene,

n-pentene, *n*-heptene, *n*-decene, acetaldehyde, 2-propenal, propanal, heptanal, and ethylene oxide. Other species predicted by Milano, but not measured experimentally, included 1,3-butadiene, 2-methyl-1,3-butadiene, methanol, and ethyne. The general discrepancy between the model and experiments for *n*-dodecane may be an indication that *n*-dodecane intermediates rapidly decompose to smaller species and ultimately formaldehyde.

An investigation of the critical low temperature reactivity pathway suggests an interesting mechanism that, if highly active, would yield high concentrations of formaldehyde. The following pathway, illustrated in Fig. S19, only includes reactions from the Livermore model. If the 2-, 3-, 4, or 5-dodecyl radical is produced and followed by O_2 addition, the next step is internal hydrogen isomerization of the dodecylperoxy radical to form a dodecylhydroperoxy radical. If the hydrogen shifts from the primary carbon to the peroxy radical portion on the secondary carbon (the Livermore model does not include a hydrogen shift from the primary carbon for 6-dodecyl radical), then the traditional low temperature reactivity pathway will always yield formaldehyde. This is because the subsequent step is O_2 addition to the alkyl radical portion on the primary carbon of dodecylhyderoperoxy radical to form peroxydodecylhydroperoxy radical, followed by cleavage of the peroxy bond attached to the secondary carbon to yield a hydroxyl radical, plus a hydrogen shift from hydrogen on the secondary carbon to the peroxy radical portion attached to the primary carbon, yielding a ketohydroperoxide. In the Livermore model, only one decomposition reaction is included for each ketohydroperoxide isomer. A hydroxyl radical is yielded from cleavage of the peroxy bond on the primary carbon, and cleavage of the bond between the carbon adjacent to the primary carbon and the carbon two removed from the primary carbon occurs, yielding formaldehyde and either a C_{11} alkanoyl radical (if the 2-dodecyl radical was produced) or else a C_{11} oxoalkyl radical isomer. C_{11} alkanoyl radical decomposes to CO and a C_{10} alkyl radical; C_{11} oxoalkyl radical decomposes to, depending on either of the three oxoalkyl isomers, C_9 alkyl radical and ketene, or C_8 alkyl radical, CO and ethene, or C_7 alkyl radical, ketene and ethene.

From the four ketohydroperoxides, the CH_2O:CO yield is 2:1, correlating with the observation that CH_2O was observed significantly greater than CO. Considering the lean conditions of these experiments, the C_7–C_{10} alkyl radicals are likely to undergo a branching path similar to that of the C_{12} alkyl radical, and thus the path leading to more formaldehyde may occur again. O_2 consumption levels suggest that 5 ppm O_2 was consumed for 1 ppm *n*-dodecane, implying that several layers of oxidation occurred. Further supporting this suggested path, ethene was the dominant alkene observed, at 640 and 700 K, and although ketene was not observed, acetic acid was measured significantly and was a major intermediate from ketene oxidation in a subatmospheric static reactor at 553–713 K [35]. In general, the measured intermediates suggest that in the critical low temperature branching pathway for *n*-dodecane, the key reaction path may be hydrogen abstraction from a secondary carbon followed by molecular oxygen addition and then internal hydrogen shift from the primary carbon.

5. Conclusions

The intermediate species produced from *n*-dodecane oxidation at lean, dilute mixtures and elevated pressure in the low temperature regime in a pressurized flow reactor were measured. These speciation results are unique in that they include temperatures as low as the NTC regime for a prototypical jet fuel component. While many species were measured in trace amounts, the results indicated carbon oxides (CO and CO_2), light aldehydes (formaldehyde, acetaldehyde, and propanal), C_{12} alkylated tetrahydrofurans (2-propyl-5-iso-pentyltetrahydrofuran and two other incompletely identified isomers), lactones (5-methyl-dihydro-2(3*H*)-furanone and dihydro-2(3*H*)-furanone), and alkenes (ethene and 3-dodecene) are sufficient to capture at least 55% of the *n*-dodecane decomposition throughout the low temperature regime. The Livermore and Milano models were compared to the results and captured most trends. In the low temperature branching pathway from Livermore, a reaction path was identified (hydrogen abstraction from a secondary carbon followed by molecular oxygen addition and then internal hydrogen shift from the primary carbon) which would, if more active, predict the measured intermediate species. Ultimately, the experiments provide benchmark data for the refinement of *n*-dodecane and surrogate jet fuel models.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data (entire reaction intermediate speciation data set, separated by class, for three *n*-dodecane experiments conducted in the pressurized flow reactor at Drexel University and additional rate-of-production analysis graphs (Figs. S1–S18) for the two models and a low tem-

perature pathway sketch (Fig. S19)) associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.038.

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